

From sample preparation to clinical confidence: quality and harmonization strategies for reliable metabolomics

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What we are going to cover

Quality

Evidence to show that an untargeted LC-MS dataset is fit for its intended use

Harmonization

Support **scomparisons** and **trust** across runs, studies and laboratories



Clinical confidence

What QA/QC can establish to make data robust and defensible in a clinical context

Quality assurance and quality control

Quality assurance

Activities that build confidence that quality requirements will be fulfilled

- Training, SOPs, equipment qualification and maintenance
- Accountability, records and periodic review
- Corrective actions

Before

Quality control

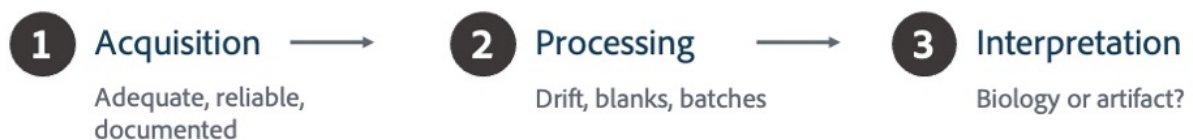
Measurements that demonstrate whether quality requirements have been met

- SST and blanks
- Pooled QC and internal standards
- Reference materials
- Data quality metrics and review

During and after

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QA/QC covers the whole workflow



Good QA/QC prevents false discoveries, protects valuable samples, and makes results reproducible across time and laboratories.

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QA/QC in untargeted metabolomics

Targeted analyses

- Defined set of metabolites
- Quantitative



Classical (bio)analytical
validation

Untargeted analyses

- Broadest coverage, non-selective
- Relative quantitation



Evaluate feature quality

Harmonisation is beneficial at every single step

- Biological question
- Suitable samples
- Experimental design and confounding factors
- Prior knowledge about the involved metabolites
- Who and how is collecting (and storing) the samples
- Duration and number of samples

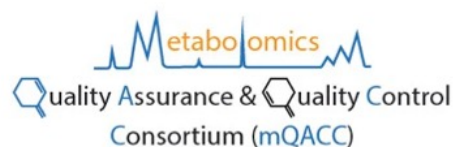
Even before measuring, it brings:

- Reproducibility
- Credibility

Metabolomics Quality Assurance and Quality Control Consortium

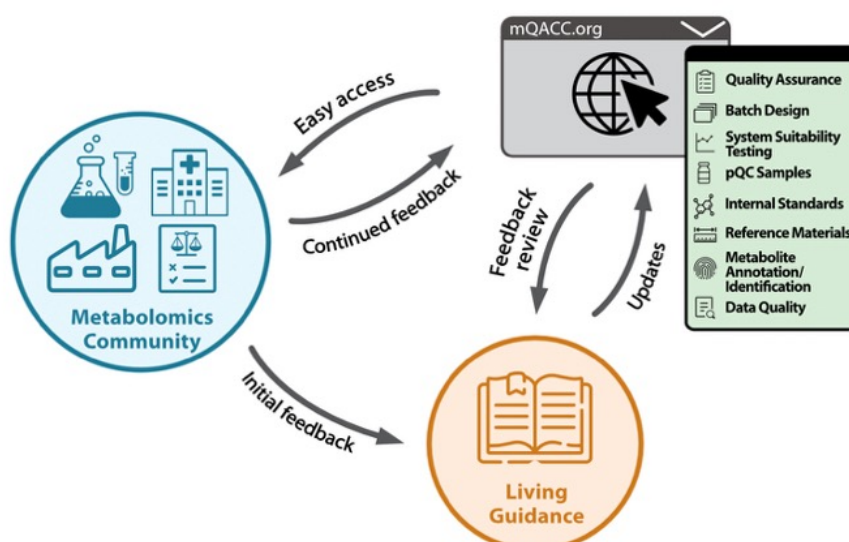
Origin and early developments

- **mQACC** was formed following a Think Tank meeting at the National Cancer Institute in October 2017
- **Mission:** To engage the metabolomics community into the development, dissemination and harmonization of best QA/QC practices in untargeted metabolomics
- **Membership:** 240 scientists from 5 continents from academia, industry and government organizations



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A Living Guidance



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Eight connected QA/QC topics

- 1

Quality assurance
- 2

Analytical batch design
- 3

System suitability testing
- 4

Pooled QC samples
- 5

Internal standards
- 6

Reference materials
- 7

Annotation / identification
- 8

Data quality review

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Minimum practices and best practices

Minimum practices

- Widely recognised by the community
- Essential baseline evidence for reliable datasets
- Documented procedures and predefined criteria
- Transparent reporting of what was done and what passed

Applicable to every study

Best practices

- Aiming at excellence
- More extensive monitoring and characterization
- Long-term and inter-laboratory comparisons
- Higher traceability and oversight
- Electronic systems and automation

A higher level of control when the application requires it

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Living Guidance #1: Quality Assurance

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Quality assurance

- Every measure taken to **ensure future analyses will be reliable**
- Instrument calibration and maintenance
- User training and proficiency testing
- Standard operation procedures
- Record keeping
- ...

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Sample preparation begins with sample provenance

Stage	Information that must remain interpretable
Collection	Matrix, tube or additive, collection time and relevant clinical context
Processing	Time and temperature before processing, centrifugation and extraction procedure
Storage and transport	Aliquot identity, storage conditions and freeze-thaw history
Preparation	Input amount, extraction batch, operator, reagents and deviations

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Define the research question and key elements of the study beforehand

- Biological phenomenon
- Experimental design and confounding factors
- Sample collection
- Sample preparation
- Analytical platform(s)
- Data analysis strategy
- And many others, depending on the study...



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Experimental design

- Fit the biological question
- Timeline
- Samples per group
- Confounding factors



Quality assurance

Minimum practices

- Maintain a documented, fit-for-purpose QA system
- Assign responsibility and maintain training records
- Schedule maintenance, calibration and suitability checks
- Record failures, corrective actions and periodic review

Best practices

- Use an appropriate standards framework
- Track sample custody and assess procedural risks
- Use audits, proficiency testing and electronic records where appropriate
- Define retention, backup and recovery procedures

A quality management system should scale to the intended use and laboratory environment.

Living Guidance #2: Analytical batch design

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Different types of samples used in QA/QC

- **System Suitability Test samples:**
 - Analysis(es) performed before the analytical batch. Check if analytical system is working appropriately and fit-for-purpose
- **Blanks:**
 - Define extraneous signals.
- **Standard Reference Materials:**
 - Well characterized, homogenous and stable.
- **Pooled QCs:**
 - Made from pooled biological samples (discussed in detail later)

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A sequence to support quality

Sequence stage	Purpose and decision
SST and initial blanks	Verify the system and method, then decide whether to proceed. Assess carryover and contaminants.
Matrix conditioning	Ensure stable operation before study samples
Study samples with pQCs	Randomized and balanced blocks with QC measurements throughout
Closing pQCs, SST, blanks	Assess end-of-batch performance: instrument performance, carryover, assist annotation/identification

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Blanks reveal different sources of contamination

System blank

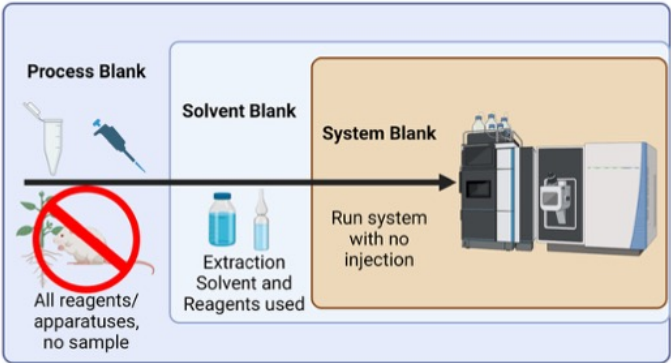
Mobile phase and LC-MS system, with no injection

Solvent blank

Injected solvent, system background and carryover

Process blank

Reagents and apparatus used through sample preparation, without biological sample



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Figure credit: Ryland Giebelhaus

Batch design

The set of experimental and technical samples from a study that are processed at one time and analyzed in a single, non-stop instrumental run

Consider some constraints:

- Total number of samples: sample, sample preparation, method, instrument, staff
- Layout: avoid bias, allow for QA and QC practices
- Multi-batch studies: additional anchoring point needed



Batch design: size

Sample/Sample prep

- How dirty are the samples?
- How many samples can be prepared in parallel?



Limiting factors



LC-MS

- Total runtime of analysis/batch
- Stability of samples in autosampler
- Stability of instrument calibration
- Staff schedule



Practical approach

- Size is assay- and instrument-dependent
- Determined batch size during method development:
 - Check signal loss during sequence
 - Apply a pre-defined cut-off (e.g., 20% loss in signal intensity)
 - Consider sample preparation and staff availability limitations



Batch design: layout

- **System suitability** samples and **blanks** at the beginning of the sequence must be inspected "**on the fly**" before the analysis of real samples starts
- **Randomization** and **balancing** of experimental groups is crucial to avoid bias
- QCs for metabolite **annotation** can be added at the end
- System suitability samples and blanks at the end of the sequence to check system performance and **carry-over**.

System suitability samples and blanks
Conditioning QC samples (5-15)
QC
QC
5-6 randomized samples
QC
5-6 randomized samples
QC
5-6 randomized samples
QC
QC
QC dilution series
QC samples for MS/MS and CCS data acquisition
System suitability samples and blanks

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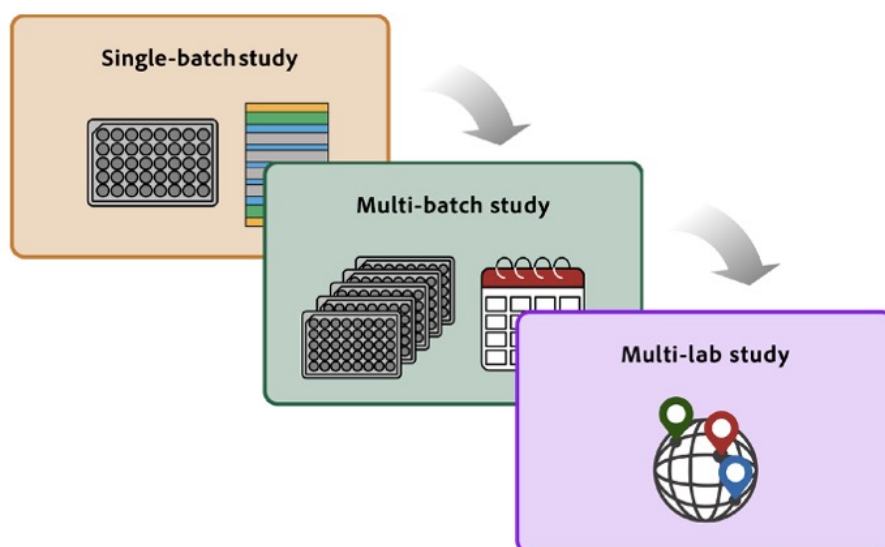
Batch design: layout

- **Double QCs** at the beginning/end of the sequence to ensure performance of drift correction algorithms
- Optimal number of **Conditioning** samples depends on sample, sample prep and method
- **Long-term reference** materials can be added to the blocks of real samples if composition not remarkably different; otherwise, at the beginning/end.
- Samples of different nature (blanks, different sample type or sample preparation) must not be injected between actual samples, as the system will have to be **reequilibrated**

System suitability samples and blanks
Conditioning QC samples (5-15)
QC
QC
5-6 randomized samples
QC
5-6 randomized samples
QC
5-6 randomized samples
QC
QC
QC dilution series
QC samples for MS/MS and CCS data acquisition
System suitability samples and blanks

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Batch design: layout



System suitability samples and blanks
Conditioning QC samples (5-15)
QC
QC
5-6 randomized samples
QC
Long term QC / SRM
5-6 randomized samples
QC
5-6 randomized samples
QC
QC
QC dilution series
QC samples for MS/MS and CCS data acquisition
System suitability samples and blanks

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Analytical batch design

Minimum practices

- **Randomize** run order and use class-balanced blocks where appropriate
- **Bracket** study samples with representative QC samples
- **Condition** the system using the study matrix
- Include solvent and process **blanks** and assess their effect on conditioning

Best practices

- Place suitable blanks at the beginning and end
- Establish batch size during method development
- Include long-term QC in large studies
- Use shared controls for comparisons across studies and laboratories

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Living Guidance #3: System Suitability Testing

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System suitability: go/no go

System suitability test: Analyses performed before the analytical batch of the study samples to check that the analytical system is working appropriately and fit-for-purpose

Universal (instrument) SST

- Calibration
- Contaminants
- Sensitivity
- Drift
- Pump pressure
- Compare against longitudinal data

Is system performing well?

Assay-specific SST

- Retention times and other metrics meet pre-defined acceptance criteria
- Verify chromatographic resolution of critical pairs
- Check extraction blank for contaminants
- Check derivatization, if applicable

Is assay performing well?

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System suitability: go/no go

Verify the instrument and method before valuable study samples are committed.

Predefined checks

- Retention time and peak shape
- Mass accuracy and resolving power
- Signal intensity
- MS/MS performance
- Pressure, contamination, carryover



PASS

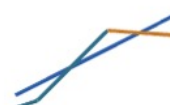
Proceed

FAIL

Correct before run

BEST PRACTICE

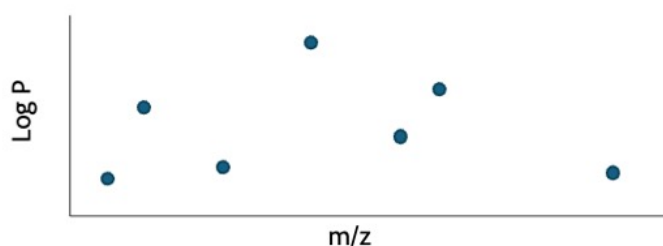
Keep track of SST results over time



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System suitability: choosing a SST mixture

- Commonly 6-10 standards used
- Span over the elution profile and m/z range of method
- Ionize in both pos and neg polarity or use different SST mixtures for each polarity
- Chemical diversity and multiple compound classes
- Plot $\log P$ vs m/z
- Cheap and readily available over time



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System suitability testing

Minimum practices

- Describe the SST material, composition and storage
- Measure LC, MS and contamination indicators
- Apply predefined acceptance criteria
- Review results before starting the analytical batch

Best practices

- Characterize stable SST material over time
- Establish acceptance limits from repeated measurements
- Track historical performance and recurrent failures
- Consider low, middle and high concentration checks

SST results determine whether the instrument and method are ready for study samples.

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Living Guidance #4: Pooled QC samples

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Types of QC samples

Pooled QC

A matrix-matched QC sample prepared by mixing aliquots from all the samples of a study. In large studies: can be prepared from a representative subset of the samples or an external source (surrogate).

Long-term QC

A bulk QC sample prepared from a set of samples, or a bulk sample/reference material purchased from a public source, used to assess analyses undertaken over a relatively long period for intra- and inter-laboratory reproducibility. Used to assess/correct for differences between separate studies on the same type of sample.

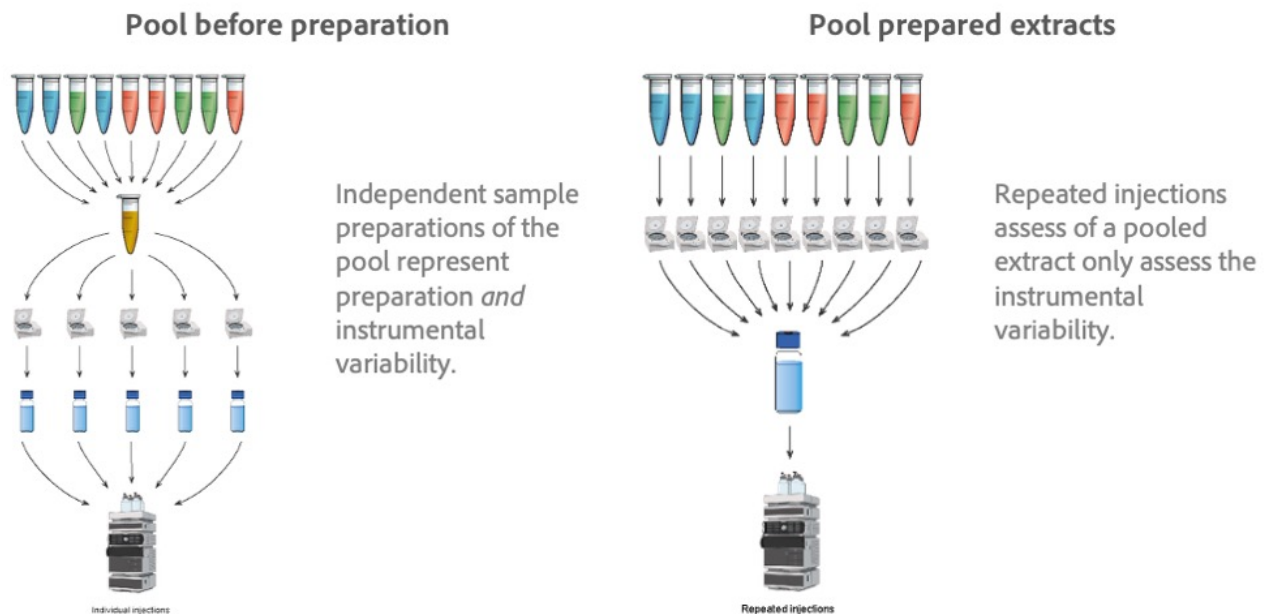
System Conditioning QC

Pooled QC or other bulk QC samples injected to condition the analytical platform with the same matrix as the study samples.

Annotation QC

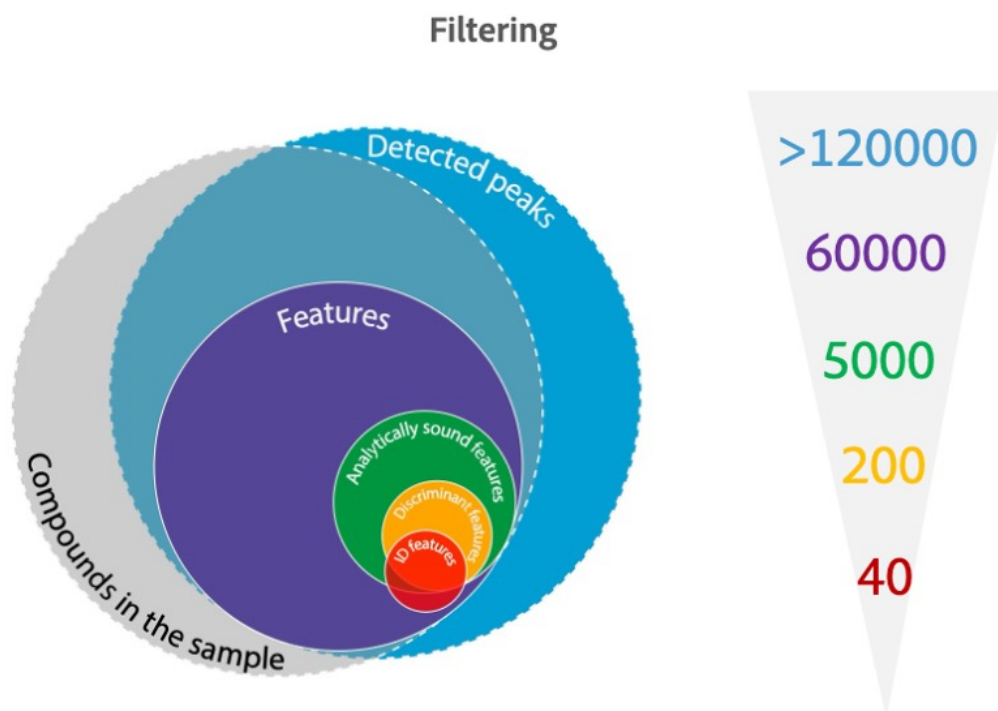
Pooled QC samples analysed, usually at the end of the sequence, to obtain in-depth, additional data for compound annotation/identification (MS/MS, CCS...)

Pooling and replication define the variability being measured



Uses of QC samples

- Evaluation of instrument/method stability
- Feature filtering
- Batch acceptance/rejection
- Evaluation/correction of drift
- System conditioning
- Metabolite annotation
- Long-term or inter-lab comparison
- Detection of analytical outliers
- Evaluate injection order effects
- Method comparisons



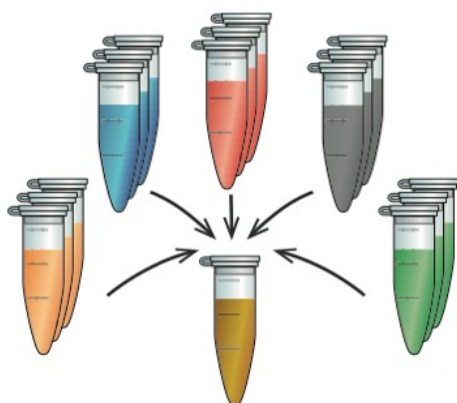
Quantitative QC metrics answer different questions

RSD	$SD(QC) / \text{mean}(QC)$ How large is repeated-measurement variation relative to signal?
D-ratio (%)	$SD(QC) / SD(\text{study samples}) \times 100$ How large is technical dispersion relative to study-sample dispersion?
Detection rate	Detected QC injections / evaluated QC injections How consistently can this feature be measured (within a group)?
Linear response to dilution	Evaluated over several concentrations/dilutions of QCs

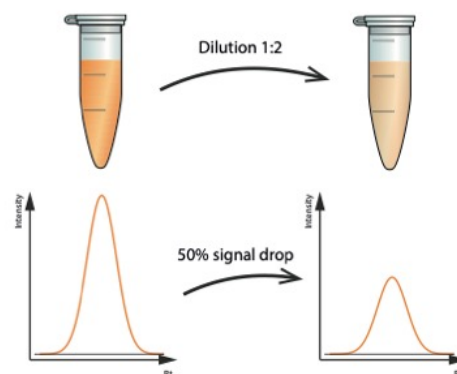
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Quality control samples (QC)

What is a QC sample?

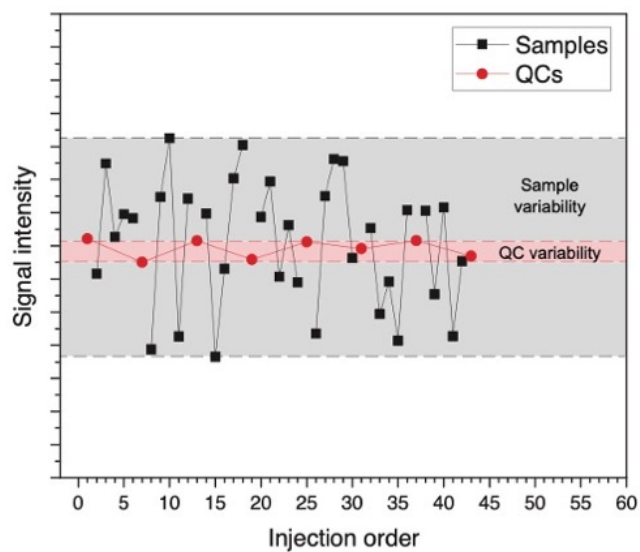


What is a diluted QC sample?



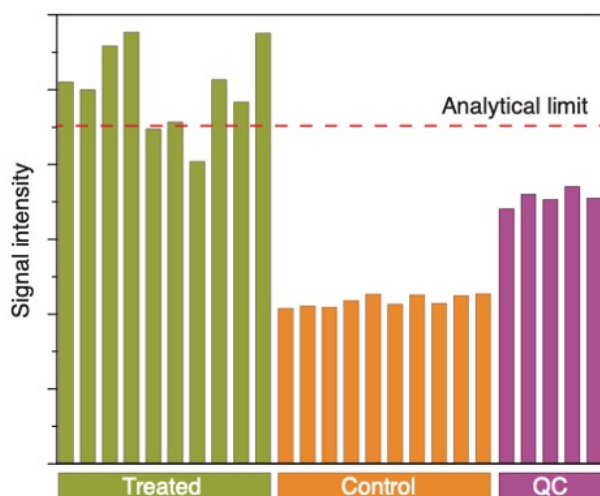
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Filtering: variability in QCs



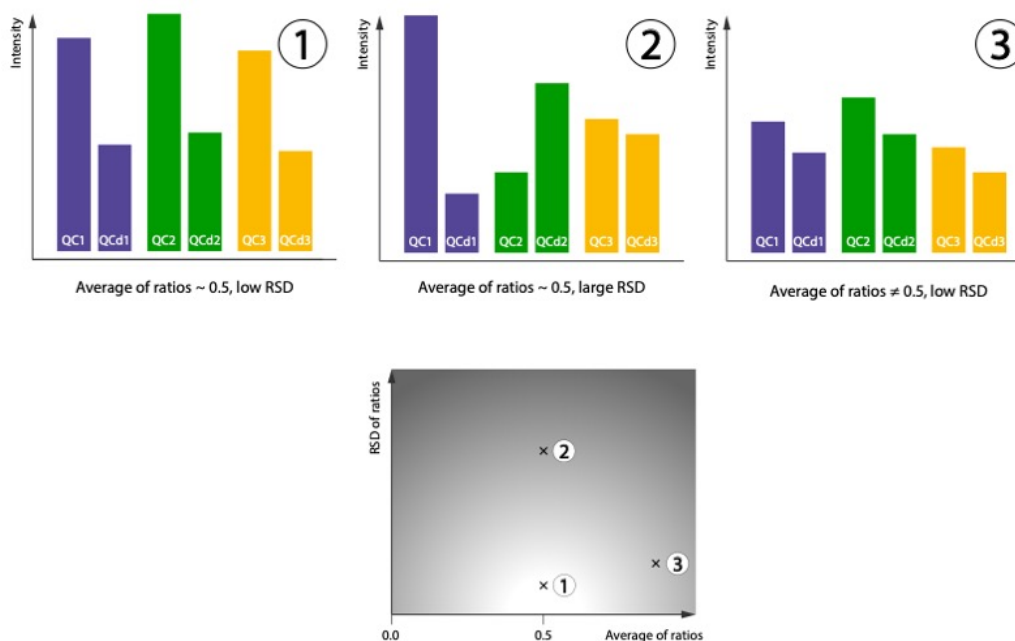
%CV of QCs must be equal to or lower than that of real samples

Filtering: presence/absence in groups



Presence in at least 80% of samples of at least 1 group

Filtering through dilution factor



Pooled QC samples

Minimum practices

- Use a representative QC sample throughout the study
- Assess LC and MS performance before, during and after acquisition
- Quantify feature-level technical repeatability
- Use predefined criteria for batch and feature acceptance

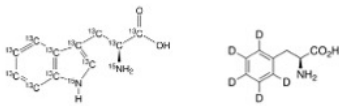
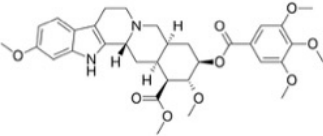
Best practices

- Use dilution or concentration series for response assessment
- Consider phenotypic pools for poorly represented analytes
- Monitor long-term QC performance
- Use suitable QC data for correction and feature filtering

Living Guidance #5: Internal Standards

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Internal standards: selection

Type	Characteristics	Number
Isotopically-labelled compounds	- Retention time - Ionization mode - m/z - Chemical diversity - Availability/Cost - Sample type/matrix - Purity	<i>Method- (and -availability) dependent!</i>
		
Exogenous compounds		
		
Isotopically-labelled extracts		

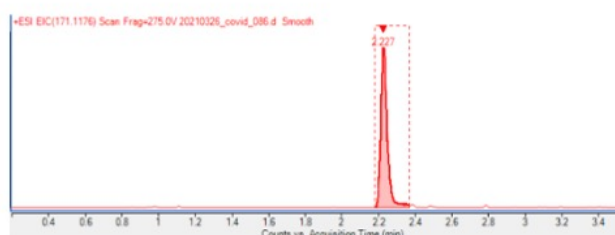
They must represent the compounds whose variability must be assessed. Increasing number of commercial options available

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Internal standards: metrics for evaluation

- Retention time
- Peak shape
- Intensity
- MS & MS/MS spectra
- Reproducibility over samples

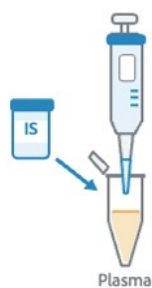
Compare to previously established cut-off criteria!



Internal standards: uses

When to spike IS?

IS before extraction



Sample

Extraction and instrumental variability



Extraction

IS after extraction



Pre-injection

Instrumental variability

Both can be combined

Internal standards

Minimum practices

- Report identity, supplier and catalogue number
- Report concentration, purity and intended purpose
- Specify isotope labels and labeling level where relevant

Best practices

- Cover relevant retention times, polarity and ion modes
- Optimize selection and concentration for each matrix
- Monitor standards during acquisition
- Check interference, stability and carryover

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Living Guidance #6: Reference Materials

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Reference materials

- Well-characterized matrix or standard mix
- Reproducible, robust
- Available in enough amount as to guarantee long-term availability
- Cheap

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Complementary roles of QC and reference materials

Material	Main role	Principal limitation
Study pooled QC	Represents the current study and tracks analytical repeatability	Composition depends on the study and pooling strategy
Long-term reference QC	Links batches or studies over an extended period	Requires sufficient supply, stability and controlled handling
Certified reference material	Supports comparison with specified certified properties	Certification covers the stated properties, not every untargeted feature
Defined standard mixture	Tests selected chromatographic or MS properties	Limited representation of the biological matrix
Internal standards	Check individual sample behaviour: Extraction, injection, matrix effects	Availability: actual representativeness of metabolites of interest

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Reference materials

Minimum practices

- If used, document material, source, lot and purpose
- Prepare, aliquot and store under controlled conditions
- Protect homogeneity and stability
- Evaluate with predefined acceptance criteria

Best practices

- Match the matrix when the purpose requires it
- Characterize in-house materials and plan sufficient supply
- Include suitable material across batches and laboratories
- Retain and share RM data with the study

A reference material provides a common comparison only for the properties it can support.

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Living Guidance #7: Metabolite annotation and identification

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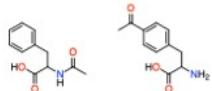
Identification requires evidence matched to an authentic compound

Reported status	Evidence in the manuscript	Interpretation
Identified	At least two independent, orthogonal data types relative to an authentic compound under identical conditions	High-confidence assignment supported by the stated comparison
Annotated	An assignment that does not meet the identification criterion	Report the confidence level, evidence and competing candidates
Not annotated or identified	No metabolite structure can be reported	Retain the analytical feature identity without inventing a chemical name

Example orthogonal evidence: retention time and MS/MS. State the confidence framework used.

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Metabolite identification Confidence levels

4	Exact mass, unequivocal molecular formula HRMS	$C_{11}H_{13}NO_3$ 207.0895 Da
3	Compound class Additional compound properties	
2	Probable structure MS fragmentation, external libraries, theoretical prediction	
1	Confirmed structure vs. reference standard, in-lab	 N-acetylphenylalanine
0	Confirmed structure At the isomer level	 N-acetyl-L-phenylalanine

Sumner et al., 2007. *Metabolomics*, 3(3), 211–221 and Blaženović, et al., 2018. *Metabolites*, 8(2), 31–23.

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Metabolite identification Confidence levels

Level	Atoms	Atomic connections	Relative stereochemistry	Chirality	Metabolite ID class	Metabolite information level
A	✓	✓	✓	✓	A single, defined enantiomer or a single, defined achiral metabolite	Known molecular formula, known structure, known stereochemistry if present, known chirality if present e.g. tartaric acid (2R,3R)-2,3-dihydroxybutanoic acid
B	✓	✓	✓	X	One of two enantiomers	Known molecular formula, known structure, known relative stereochemistry if present unknown chirality e.g. tartaric acid: enantiomer undefined (2R,3R or 2S, 3S)
C	✓	✓	X	X	One of a number of stereoisomers e.g. E/Z geometric or cis-/trans- ring isomers	Known molecular formula, known molecular structure, unknown relative stereochemistry e.g. tartaric acid diastereomer undefined: 2R, 3S, 2R, 3R or 2S, 3S oleic acid (Z-isomer) or elaidic acid (E-isomer of 9-octadecanoic acid)
D	✓	X	X	X	One of a number of positional isomers	Known molecular formula, known functional groups, unknown structure e.g. C ₁₈ H ₃₄ O ₂ : 9-octadecanoic acid or 7-octadecanoic acid
E	✓	X	X	X	One of a number of possible compounds of known molecular formula	Known molecular formula, unknown structure e.g. C ₂ H ₆ O: dimethylether or ethanol
F	X	X	X	X	Specific spectral features defining a structural class	Unknown molecular formula known structural class
G	X	X	X	X	Specific spectral features	Unknown molecular formula

http://metabolomicsociety.org/images/newsletters/Revised%20Structure-Based%20Reporting%20Standards%20for%20Metabolite%20Identification_141119.pdf

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Metabolite annotation and identification

Minimum practices

- Report acquisition, processing and matching methods
- State the confidence system and evidence for each assignment
- Report ambiguous candidates and unassigned signals
- Manually verify biologically important assignments

Best practices

- Combine independent lines of chemical evidence
- Share the experimental evidence used for assignments
- Report structure identifiers, software and database versions
- Check biological plausibility and define acceptance criteria

Chemical confidence limits the reliability of the biological interpretation.

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Living Guidance #8: Data quality review and reporting

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Data quality review

Must precede data analysis: evaluate the batch first, then features, then the integrated multi-batch dataset.

- 1** **Batch acceptance** Predefined metrics; accept, investigate or reject
- 2** **Outlier detection** Study samples and QC samples
- 3** **Feature filtering** Detection rate, QC-RSD, D-ratio, linearity
- 4** **Correction** Within- and between-batch drift
- 5** **Documentation** Decisions, causes and corrective actions

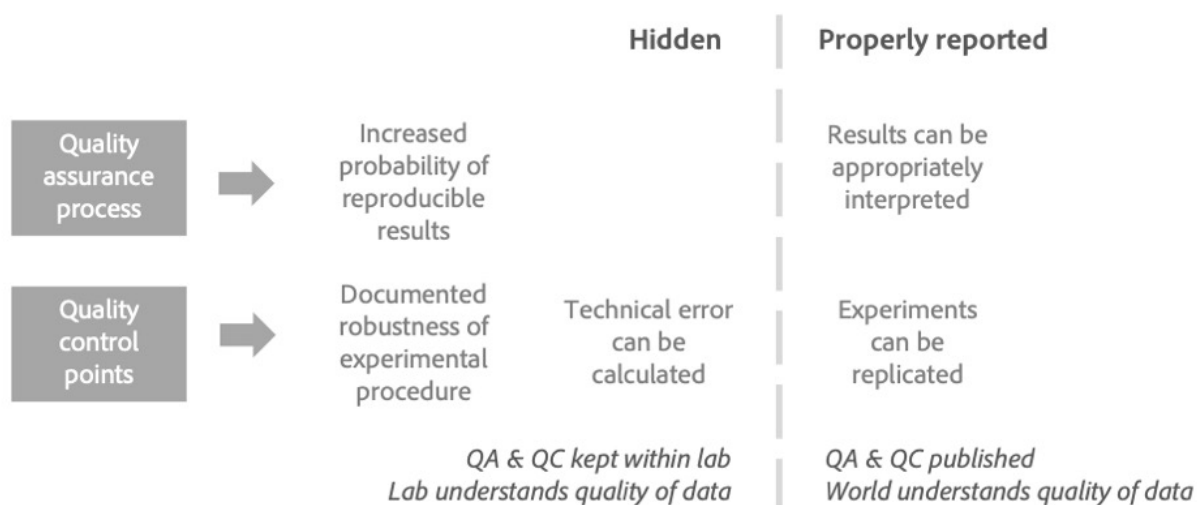
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Quality review leads to explicit decisions

Decision level	Evidence	Possible documented response
Before acquisition	SST and contamination checks	Proceed, investigate or repeat SST
Batch or batch segment	QC repeatability, drift and predefined criteria	Accept, hold or reanalyze the affected interval
Individual sample	IS behavior, injection record and outlier criteria	Review, reinject, reprepare or exclude with a reason
Individual feature	Detection, blank contribution and repeatability	Retain, correct if supported or filter
Combined study	Residual batch effects and reference performance	Release for interpretation or investigate further

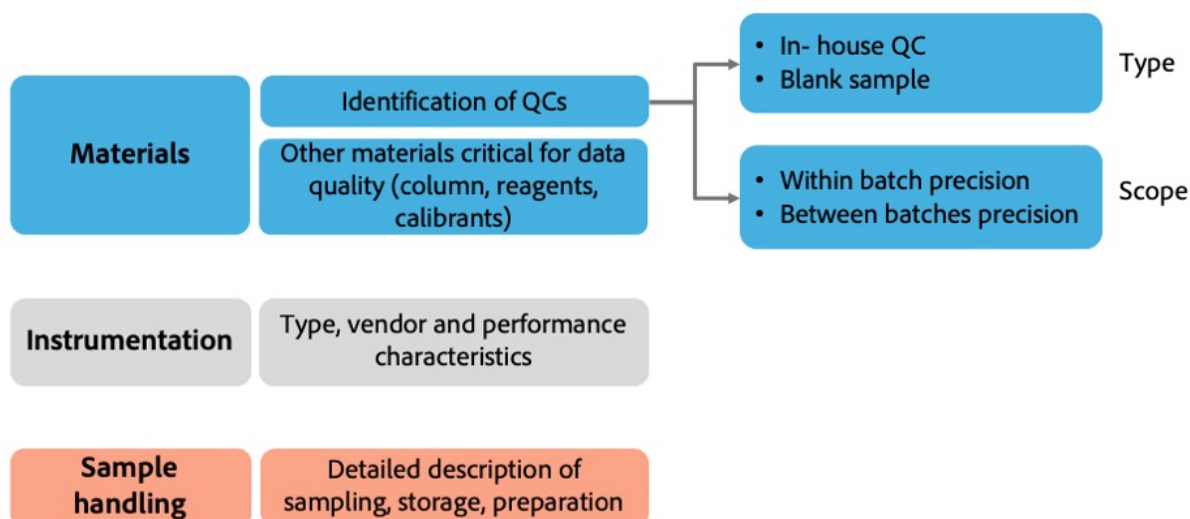
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Data quality review: avoid "hidden quality"



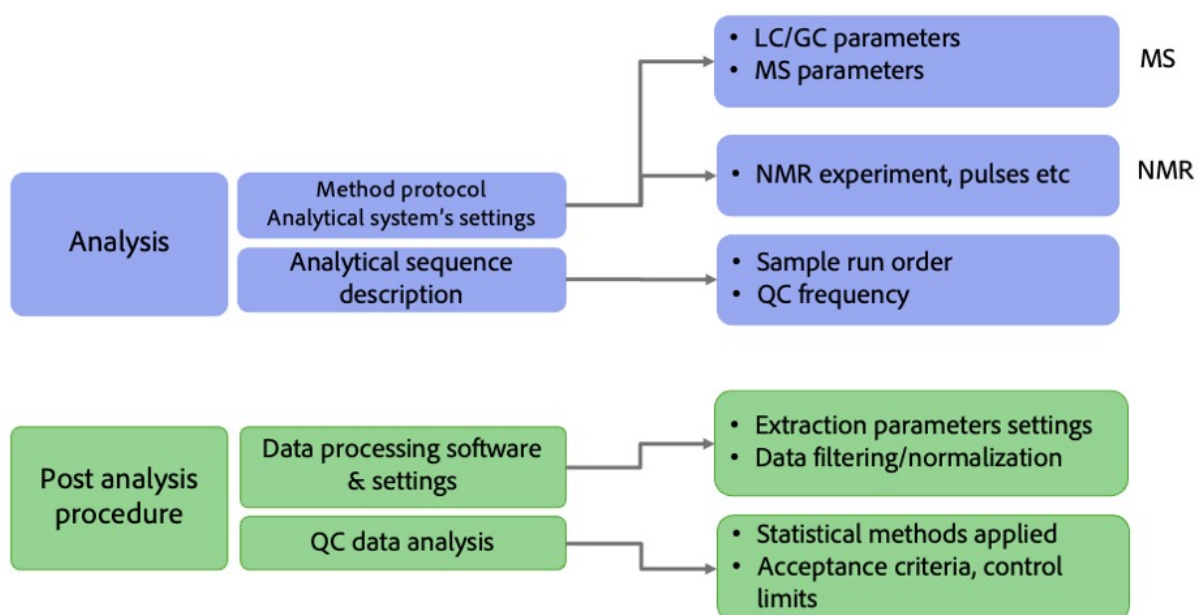
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What to report?



Slide credit: Mariana Silva

What to report?

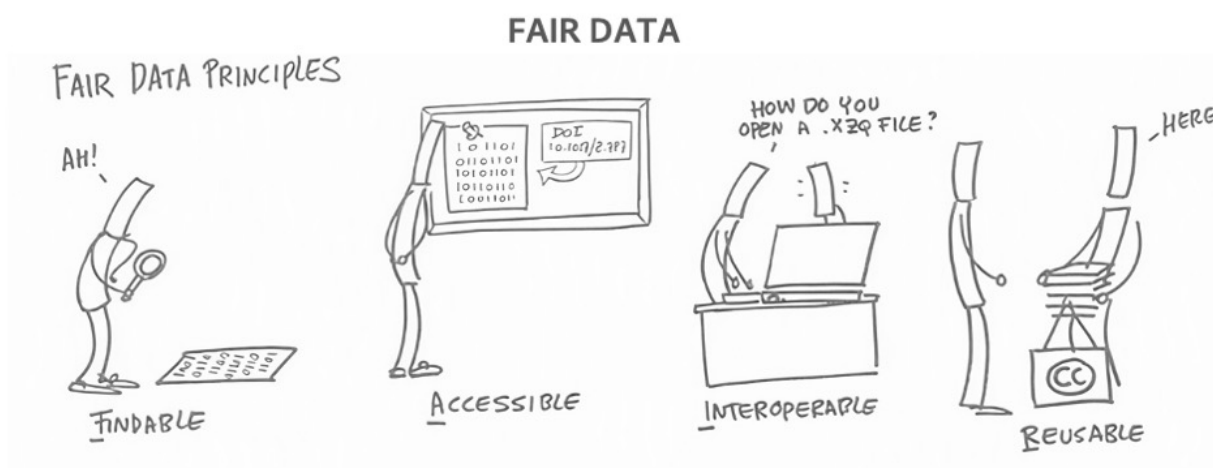


Slide credit: Mariana Silva

The reporting package makes quality assessable

Materials and sequence	QC preparation, source and purpose, quantity, frequency, run order and blanks
Performance and decisions	Acceptance criteria, measured metrics, failed samples or batches, and follow-up actions
Processing and assignments	Software, parameters, correction, filtering, chemical evidence and confidence levels
Reusable evidence	Raw and processed data, QC and RM data, metadata and versioned records

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Findable
Accessible

Structured metadata
In proper data repositories

Interoperable

Standard formats
(mzML, nmrML, ISA-Tab)

Reusable

Public repositories
Complete documentation

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Data quality review

Minimum practices

- Assess each batch before statistical or biological analysis
- Use predefined batch and sample acceptance rules
- Evaluate inter-batch effects after individual batches pass
- Apply justified correction and feature filtering

Best practices

- Monitor selected indicators during acquisition
- Set triggers for corrective action or stoppage
- Compare with historical performance
- Investigate failures and improve the process

A complete review considers batch quality, individual samples and individual features.

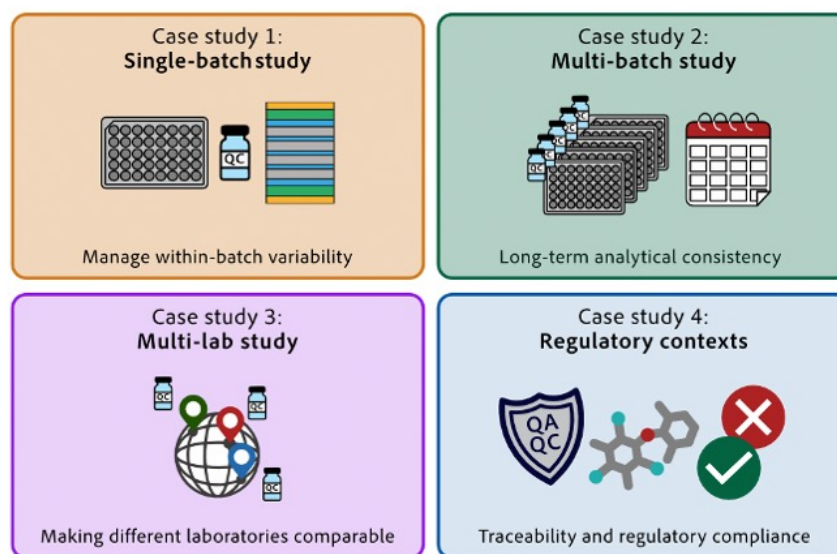
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Case studies

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Four study settings organize the practical application

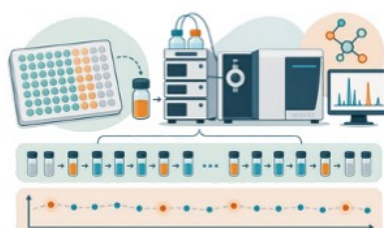


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Case 1. A single-batch clinical research study

Study context	A case-control plasma study measured in one continuous LC-MS run.
Main risk	An apparent difference between groups could reflect run order, preparation variability or contamination.



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Case 1. Evidence for a defensible batch decision

Design	Balance groups through preparation and injection order. Record both sequences.
Monitor	Review SST, blanks and bracketed pQCs. Use appropriate IS signals for individual samples.
Decide	Apply predefined criteria to the batch, samples and features before comparing groups.
Document	Retain deviations, follow-up actions and the final set of accepted measurements.

Outcome: a defensible dataset with controlled within-batch variability.

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Case 2. A study distributed across many batches

Study context	A hospital cohort is analyzed over several months.
Main risk	Instrument drift, maintenance, sample collection...



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Case 2. A bridge across time

Before the study	Plan sufficient representative pQC and stable long-term QC or RM material.
Within each batch	Apply the same documented design and batch acceptance process.
Across batches	Compare shared controls, historical performance and residual batch effects.
After correction	Evaluate improvement and document any remaining limitations.

Outcome: integrated batches that remain comparable over time.

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Case 3. A study involving several laboratories

Study context	Several hospitals or core facilities contribute metabolomics measurements.
Main risk	Site differences can arise from preparation, methods, feature detection or chemical assignments.



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Case 3. Harmonization requires shared evidence

Align procedures	Agree on sample handling, preparation, QC purpose and essential metadata.
Use common materials	Distribute suitable stable materials and evaluate site performance.
Compare assignments	Match chemical evidence and confidence, with transparent treatment of ambiguity.
Define the common dataset	Retain measurements that meet the intended comparability criteria.

Outcome: transparent comparability without hiding laboratory effects.

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Case 4. A study in a regulated or clinical laboratory setting

Study context	An untargeted workflow operates within a formal laboratory quality system.
Main risk	A quality claim may be difficult to defend without controlled decisions and traceable records.



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Case 4. The record supporting release

Evidence	A traceable record
Sample and method	Identity, custody, SOP version, preparation history and deviations
System and controls	Qualification, maintenance, calibration, SST and QC results
Data and assignments	Original files, processing version, exclusions and identification evidence
Review and authorization	Acceptance criteria, reviewer, failure investigation and release decision
Retention and change	Secure records, backup, change assessment and follow-up

Outcome: reproducible evidence suitable for audit and regulatory review.

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Wrap-up

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How the evidence expands across study settings

Setting	Main additional challenge	Evidence with particular value
Single batch	Within-run variability	Bracketed pQCs, blanks, sample-level checks and predefined review
Multiple batches	Consistency over time	Shared long-term QC or RM, change history and inter-batch evaluation
Multiple laboratories	Comparability between settings	Common materials, agreed metadata and comparable chemical evidence
Regulated context	Traceability and defensible decisions	Controlled procedures, complete audit trail and authorized release

All four settings use the same eight topics. Their risks determine the depth of implementation.

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Analytical quality and clinical confidence

The guidance supports

- Reliable analytical measurements within a defined scope
- Transparent chemical assignments
- Documented comparison across relevant batches and laboratories

Clinical use also needs

- A clearly defined intended clinical use
- Application-specific analytical and clinical validation
- Appropriate interpretation and authorization within the laboratory system

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An implementation plan for the next study

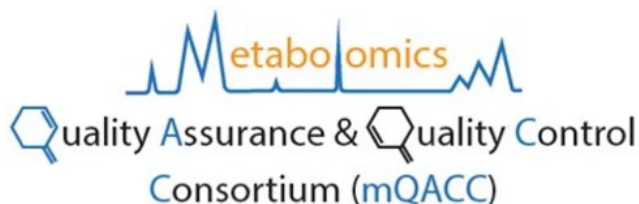
Design and ownership	Define intended use, accountable reviewers, SOPs and acceptance criteria.
Materials and sequence	Plan representative controls, supply, stability, randomization and batch structure.
Review and reporting	Specify real-time triggers, batch review, feature handling and the reporting record.
Learning from failures	Review causes, document changes and evaluate their effect on later performance.

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Where to get support

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We are not alone

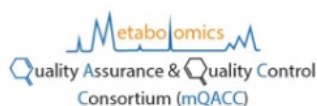


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Lan

Quality Assurance
Srujana Golla & Shaowei
An

*What is "best
practice"*

*What is
"common
practice"*

*What do we
need to
report?*

*How can we
improve
practices?*

*How do we
adapt to new
technologies?*

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mQACC outputs

- Workshops
- Oral and poster presentations
- Publications



Quality Assurance and Quality Control Reporting in Untargeted Metabolic Phenotyping: mQACC Recommendations for Analytical Quality Management
J Kirwan et al 2022 Metabolomics



Current practices in LC-MS untargeted metabolomics: a scoping review on the use of pooled quality control samples
Broeckling CD et al 2023 Analytical Chemistry



Procedural Rigor and Reproducibility in NMR Metabolomics: Community Practices and Challenges
Cheng LL et al, 2026 Crit Rev Anal Chem



Best Practices in GC-MS and GC x GC-MS-Based Metabolomics and Volatile Analyses: An International Survey
Giebelhaus RT et al, 2026 Analytical Chemistry

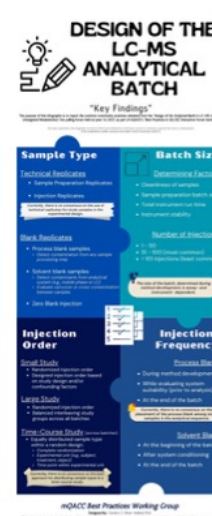
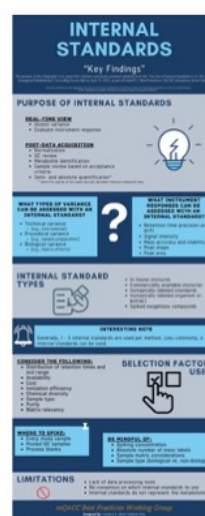
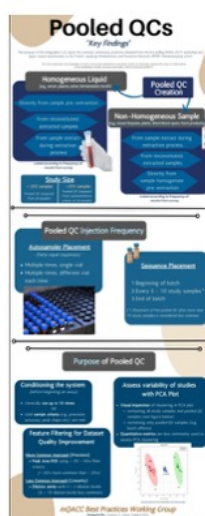


Establishing a framework for best practices for quality assurance and quality control in untargeted metabolomics
Mosley JD et al 2024 Metabolomics

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mQACC outputs

- Workshops
- Oral and poster presentations
- Publications
- Living Guidance Infographics
- Living Guidance (coming soon!)



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BP4NTA



<https://nontargetedanalysis.org/>

- List of software tools
- Useful Links
- NTA SRT



NTA Study Reporting Tool (SRT)

Want to use the Study Reporting Tool to guide your study design, evaluate your own manuscript, or conduct a peer-review?

You're in the right place! You can download Excel or PDF versions of the SRT on this page. The image below shows a preview of the SRT. In the downloadable versions (available via links above the image), you can use the Scoring dropdown menus to assign color/numeric scores and use the Rationale column to explain your selection. The Excel file also has a tab in which you can easily plot scores from multiple reviewers.

Download the SRT as an Excel file

Note: The Excel file includes the SRT, a plotting functionality to easily view scores from multiple reviewers, and a table that explains the scoring levels.

Download the SRT as a PDF file

Note: The scoring drop-down colors will not work when viewing the PDF in an Internet browser or in Mac Preview – please use a PDF application such as Adobe.

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Take home messages

- 1** Design quality Experimental design, QA, batch design and predefined criteria.
- 2** Use layered evidence SST, blanks, pQC, internal standards and RMs
- 3** Review before interpreting Accept batches and features before testing biology.
- 4** Scale fit-for-purpose Move from minimum to best practices as risk and context demand.

Reliable data is not assumed, but demonstrated!

Disclaimer

I am an active member of mQACC, and co-chair of the "Living Guidance" working group.

Nevertheless, mQACC has not reviewed, approved or endorsed this presentation



Fertility Metabolomics Group



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Thank you! Questions?

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